

SOY: THE HOPES PROBABLY OUTWEIGH THE FEARS

[John Livesey](#) PhD, Scientific Officer, [Department of Endocrinology](#), [Christchurch Hospital](#), New Zealand

Soy products have attracted much attention in recent years as possible beneficial additions to the traditional Western diet. The hope is that increased soy consumption will reduce the incidence of Western afflictions such as heart disease and cancer and so increase longevity. In addition, for vegetarians and vegans, soy products are a relatively cheap and convenient source of high quality protein.

There is much circumstantial evidence in the medical literature for beneficial effects of soy on human health. In relatively short-term human experimental studies consumption of soy products (as compared to animal products) is found to:

- Alter blood lipid¹ and homocysteine² concentrations in ways likely to reduce the risk of heart disease.
- Lower blood pressure in patients with hypertension³.
- Alter bone metabolism in a way that may reduce the risk of osteoporosis⁴.
- Improve memory⁵.
- Decrease menopausal symptoms⁶.

Further, observational and epidemiological studies of humans suggest that soy consumption may:

- Reduce the risk of breast cancer^{7,8}.
- Reduce the risk of prostate cancer⁹.
- Reduce the need for hysterectomy¹⁰.
- Increase bone mineral density¹¹.
- Reduce the risks of lung cancer¹² and stomach cancer¹³ in Asian populations where these diseases are relatively common.
- Assist in body weight control¹⁴.
- Reduce the risk of thyroid cancer¹⁵.

Despite this apparently weighty evidence for health benefits from soy (only a small fraction of studies with similar conclusions have been referred to), not everyone is convinced.

Several years ago an epidemiological study of elderly Japanese-Americans in Hawaii found a correlation between eating tofu in middle age and poor cognitive function, enlargement of the ventricles and low brain weight in old age¹⁶. It is not clear though what to make of these findings. They have yet to be confirmed by any other study and they appear to run counter to the finding of lower rates of dementia in Okinawa (and Japan generally) than are found in the United States¹⁷. Soy consumption is considerably higher in Okinawa than in the US. Further, two recent short-term studies suggest that soy or soy products may improve memory^{5,18}. It has been suggested that the high aluminium content of Hawaiian tofu may be responsible for the deterioration of mental function seen there. Another possibility is that the tofu consumers may have been relatively deficient in vitamin B₁₂ compared to the non-tofu consumers, who were presumably eating meat. B₁₂ deficiency can result in neurological problems.

New Zealand researchers have questioned the safety of soy for infant feeding^{19,20}, citing largely theoretical concerns about possible anti-estrogen and anti-thyroid effects from the isoflavones (phytoestrogens) that are found in much higher levels in soybeans than in common Western foods. However a recent study of 248 young adults who

were fed soy formula as infants found no statistically significant differences between them and 563 young adults fed cow milk formula as infants²¹. With the primary aim of studying effects that might be expected from any estrogen-related activity of the soy formula, the subjects were questioned about pubertal maturation, menstrual and reproductive history, height, weight, current health and education. On this basis thirty primary comparisons were made between the two groups and using the strictly correct statistical interpretation (Bonferroni adjustment for multiple comparisons), no significant differences between the soy-fed and the cow milk-fed groups were found.

If the statistical criteria are relaxed and each of the primary comparisons considered in isolation, the only significant differences found were that women in the soy formula group had a longer duration of menstrual bleeding (by 0.4 days) and greater discomfort during menstruation²¹.

If unplanned comparisons suggested by the data are also examined, the relaxed statistical criteria find a further statistically significant difference between the two groups²¹. A greater proportion of the soy-fed group (17%) were regular users of asthma or allergy drugs than were the cow milk-fed group (10%). If however irregular users of these drugs were included with regular users, there was no significant difference between the groups for the proportions of subjects using asthma and allergy drugs. No significant difference in the incidence of thyroid disease was found for the two groups²². Thus it would appear on current evidence that feeding infants soy formula is not associated with a substantial risk to health in early adult life, though more research is needed²³, particularly with much larger numbers of (human) subjects.

This lack of obvious effect of soy isoflavones on infant development may be because humans do not absorb the isoflavones intact and only their glucuronide and sulfate metabolites are detectable in the blood²⁴.

Articles have also appeared in more popular media casting doubt (usually quite vigorously) on the merits of consuming soy (see references ²⁵ and ²⁶ for examples). Concerns in these articles centre around soy's high content compared to other plant foods of phytoestrogens and anti-metabolites such as anti-trypsin, phytic acid, hemagglutinins (lectins) and antithyroid agents. It is also claimed that soy increases the body's requirement for vitamins B₁₂ and D, and that toxic lysinoalanine, nitrosamines and monosodium glutamate are formed during the processing of soy.

The possible adverse effects of these substances in the human diet are largely speculative. An alternative view is that some compounds in soy may have desirable anti-cancer properties. Indeed, genistein exposure during puberty seems to reduce the risk of breast cancer²⁷.

At least some of these anti-soy claims appear to be weak since anti-trypsin and lectins are inactivated by heat (ie cooking)²⁸, and infants fed soy based-formula maintain similar vitamin D levels²⁹ and growth rates³⁰ to those fed cow milk-based formula.

It was reported from a retrospective study in 1990 that children with autoimmune thyroid disease (a rare condition) were more likely than their unaffected siblings to have been fed soy based-formula³¹. This finding though still awaits confirmation by other studies and the relationship found could be an example of reverse causation. That is to say, the babies may have been given soy-formula for some reason related to a predisposition to an autoimmune condition.

In 1991 a Japanese group reported that in an uncontrolled trial consumption of 30g of soybeans per day by adults resulted in half their subjects developing goiters³². However subsequent better designed studies in the USA using soy proteins failed to find such an effect or indeed any other clinically significant hormonal effects^{33,34,35}. Indeed, the claimed effect of soybeans would predict a high incidence of goiter in high soy consuming areas such as Japan and Okinawa, but this appears not to be the case. There is also evidence that soy consumption may be protective against thyroid cancer¹⁵.

It is difficult though to completely discount concerns about soy since appropriate large-scale epidemiological studies have not yet been done. What is needed are studies comparing disease and death rates for large numbers of high soy and low soy consumers living similar lifestyles. More definitive answers should be available in a few years because Professor Gary Fraser of Loma Linda University, California, has recently been funded to study the effects of soy consumption among 125,000 Seventh Day Adventists.

Allergy to soy though is well documented. For example, 10% of infants allergic to cow milk were also allergic to soy formula³⁶, but soy allergy appears to be much less common than allergy to cow milk³⁷.

The best evidence presently available on the long term effects of soy on human health is firstly the study of young adults fed soy as infants²¹ mentioned above and secondly the observation that in countries where soy is widely consumed people tend to have better longevity than most. The Japanese, regular soy consumers, lead the world in longevity³⁸. In particular, in Okinawa, where for older persons the soy consumption of about 80g per day³⁹, mainly as tofu, exceeds that in the rest of Japan, the average longevity (81.2 years) also exceeds that of the rest of Japan^{38,40}.

In conclusion, whether or not soy consumption really does benefit your health overall remains to be determined by good epidemiological studies. The balance of the evidence to date though favours a beneficial effect of soy compared to animal products. On the other hand, if it is harmful, the harm would seem not to be very great in view of the excellent longevity found in high soy consuming areas of the world.

Similar conclusions are reached in articles by [Dr John McDougall](#) and the [Vegetarian and Vegan Foundation](#).

REFERENCES

1. [Erdman JW](#). Soy protein and cardiovascular disease. *Circulation* 102:2555-9 (2000).
2. [Tonstad S, Smerud K, Hrie L](#). A comparison of the effects of 2 doses of soy protein or casein on serum lipids, serum lipoproteins and plasma total homocysteine in hypercholesterolemic subjects. *American Journal of Clinical Nutrition* 76:78-84 (2002).
3. [Rivas M, Garay RP, Escanero JF et al](#). Soy milk lowers blood pressure in men and women with mild to moderate essential hypertension. *Journal of Nutrition* 132:1900-2 (2002).
4. [Uesugi T, Fukui Y, Yamori Y](#). Beneficial effects of soybean isoflavone supplementation on bone metabolism and serum lipids in postmenopausal Japanese women: a four-week study. *Journal of the American College of Nutrition* 21:97-102 (2002).
5. [File SE, Jarrett N, Fluck E et al](#). Eating soy improves human memory. *Psychopharmacology* 157:430-6 (2001).
6. [Han KK, Soares JM, Haidar MA et al](#). Benefits of soy isoflavone therapeutic regime on menopausal symptoms. *Obstetrics & Gynecology* 99:389-94 (2002).
7. [Jakes RW, Duffy SW, Ng FC et al](#). Mammographic parenchymal patterns and self-reported soy intake in Singaporean Chinese women. *Cancer Epidemiol. Biomarkers Prev.* 11:608-13 (2002).
8. [Korde LA, Wu AH, Fears T et al](#). Childhood soy intake and breast cancer risk in Asian American women. *Cancer Epidemiol Biomarkers Prev.* 2009 Mar 24.
9. [Castle EP, Thrasher JB](#). The role of soy estrogens in prostate cancer. *Urologic Clinics of North America* 29:71-81 (2002).
[Jacobsen BK, Knutsen SF, Fraser GE](#). Does high soy milk intake reduce prostate cancer incidence? The Adventist Health Study (United States). *Cancer Causes Control* 9:553-7 (1998)
10. [Nagata C, Takatsuka N, Kawakami N, Shimizu H](#). Soy product intake and premenopausal hysterectomy in a follow-up study of Japanese women. *European Journal of Clinical Nutrition* 55:773-7 (2001).
11. [Mei J, Yeung SS, Kung AW](#). High dietary phytoestrogen intake is associated with higher bone mineral density in postmenopausal but not premenopausal women. *Journal of Clinical Endocrinology and Metabolism* 86:5217-21 (2001).
12. [Seow A, Poh WT, The M et al](#). Diet, reproductive factors and lung cancer risk among Chinese women in Singapore: evidence for a protective effect of soy in non-smokers. *International Journal of Cancer* 97:365-71 (2002).
13. [Nagata C, Takatsuka N, Kawakami N, Shimizu H](#). A prospective cohort study of soy product intake and stomach cancer death. *British Journal of Cancer* 87:31-6 (2002).
14. [Maskarinec G, Novotny R, Tasaki K](#). Dietary patterns are associated with body mass index in multiethnic women. *Journal of Nutrition* 130:3068-72 (2000).
15. [Horn-Ross PL, Hoggatt KJ, Lee MM](#). Phytoestrogens and thyroid cancer risk: the San Francisco Bay Area thyroid cancer study. *Cancer Epidemiol. Biomarkers Prev.* 11:43-9 (2002).
16. [White LR, Petrovitch H, Ross GW et al](#). Brain aging and midlife tofu consumption. *Journal of the American College of Nutrition* 19:242-55 (2000).
17. [Willcox B, Willcox C, Suzuki M](#). *The Okinawa Way: How to Improve your Health and Longevity Dramatically*. Michael Joseph, London (2001), p46. Published in the USA as *The Okinawa Program : How the World's Longest-Lived People Achieve Everlasting Health--And How You Can Too*
18. [Messina M, Gardiner C, Barnes S](#). Gaining insight into the health effects of soy but still a long way to go: commentary on the Fourth International Symposium on the Role of Soy in Preventing and Treating Chronic Disease. *Journal of Nutrition* 132:547S-551S (2002).
19. Irvine C, Fitzpatrick M, Robertson I, Woodhams D. The potential adverse effects of soybean phytoestrogens in infant feeding [Letter]. *New Zealand Medical Journal* 24 May:208-9 (1995).

20. Fitzpatrick M. Soy formulas and the effects of isoflavones on the thyroid. *New Zealand Medical Journal* 113:24-6 (2000).
21. [Strom BL, Schinnar R et al.](#) Exposure to soy-based formula in infancy and endocrinological and reproductive outcomes in young adulthood. *Journal of the American Medical Association* 286:807-14 (2001).
22. Strom BL, Schinnar R et al. Exposure to soy-based formula in infancy [Letter]. *Journal of the American Medical Association* 286:2402-3 (2001).
23. [Badger TM, Ronis MJ, Hakkak R et al.](#) The health consequences of early soy consumption. *Journal of Nutrition* 132:559S-565S (2002).
24. [Setchell KD, Brown NM et al.](#) Evidence for lack of absorption of soy isoflavone glycosides in humans, supporting the crucial role of intestinal metabolism for availability. *American Journal of Clinical Nutrition* 76:447-53 (2002).
25. [Freeman V.](#) Oriental odyssey. *Soil & Health* 61:35 (2002). This magazine now called *Organic NZ*.
26. http://www.westonaprice.org/soy/soy_alert.html
27. [Warri A, Saarinen NM et al.](#) The role of early life genistein exposures in modifying breast cancer risk. *British Journal of Cancer* 98:1485-93 (2008).
28. Dock LL, Floros JD. Thermal and nonthermal preservation methods. In: Schmidl MK, Labuza TP. *Essentials of functional foods*. Aspen, Gaithersburg, (2000) pp49-87.
29. [Hillman LS, Chow W et al.](#) Vitamin D metabolism, mineral homeostasis, and bone mineralization in term infants fed human milk, cow milk-based formula, or soy-based formula. *Journal of Pediatrics* 112:864-74 (1988).
30. [Steichen JJ, Tsang RC.](#) Bone mineralization and growth in term infants fed soy-based or cow milk-based formula. *Journal of Pediatrics* 110:687-92 (1987).
31. [Fort P, Moses N. et al.](#) Breast and soy-formula feedings in early infancy and the prevalence of autoimmune thyroid disease in children. *Journal of the American College of Nutrition* 9:164-7 (1990).
32. [Ishizuki Y, Hirooka Y. et al.](#) The effects on the thyroid gland of soybeans administered experimentally in healthy subjects. *Nippon Naibunpi Gakki Zasshi – Folia Endocrinologica Japonica* 67:622-9 (1991).
33. [Duncan AM, Underhill KE et al.](#) Modest hormonal effects of soy isoflavones in postmenopausal women. *Journal of Clinical Endocrinology and Metabolism* 84:3479-84 (1999).
34. [Persky VW, Turyk ME et al.](#) Effect of soy protein on endogenous hormones in postmenopausal women. *American Journal of Nutrition* 75:145-53 (2002).
35. [Dillingham BL, McVeigh BL et al.](#) Soy protein isolates of varied isoflavone content do not influence serum thyroid hormones in healthy young men. *Thyroid* 17:131-7 (2007).
36. [Klemola T, Vanto T et al.](#) Allergy to soy formula and to extensively hydrolysed whey formula in infants with cow's milk allergy: a prospective, randomized study with follow-up to the age of two years. *Journal of Pediatrics* 140:219-24 (2002).
37. [Nowak-Wegrzyn A, Conover-Walker MK, Wood RA.](#) Food-allergic reactions in schools and preschools. *Archives of Pediatric and Adolescent Medicine* 155:790-5 (2001).
38. [Willcox B et al.](#) *Ibid* p2.
39. [Willcox B et al.](#) *Ibid* p158.
40. [Cockerham WC, Yamori Y.](#) Okinawa: an exception to the social gradient of life expectancy in Japan. *Asia Pacific Journal of Clinical Nutrition* 10:154-8 (2001).